We may associate the above equation with a positive charge by writing it in the form

$$\frac{d\mathbf{v}_k}{dt} = \left[-\frac{1}{\hbar^2} \frac{\partial^2 E(\mathbf{k})}{\partial \mathbf{k}\, \partial \mathbf{k}} \right] |e|\mathscr{E} \qquad (5\text{-}37b)$$

By associating the absence of negative charge in the empty state k with positive charge we may think of empty state as a particle of charge + e and mass m^* given by

$$\frac{1}{m^*} = -\frac{1}{\hbar^2} \frac{\partial^2 \mathrm{E(k)}}{\partial \mathbf{k}\, \partial \mathbf{k}} \qquad (5\text{-}38)$$

Normally the empty states are found near the top of the band where the curvature of E vs. k is negative and the mass defined in Eq. 5-38 is positive. We are thus able to describe the current as due to negatively charged electrons with negative effective mass whose k-vectors are opposite to those of the empty states or we can think of the empty states as positive charges with positive mass. These latter entities, called ***holes,*** provide a convenient description of the behavior of many materials. The hole concept is widely used and accepted. In either case the charge transport is due to electron motion and not to motion of the positive ions of the lattice.

Near the top of the band an electron has negative mass and a hole has positive mass. Near the bottom of the band, however, the reverse is true, for the electrons have positive effective mass and the holes have negative mass. In an equilibrium (minimum energy) situation the electrons tend to sink to the bottom of the band and the holes tend to rise to the top. The kinetic energy of a hole is measured downward from the top of the band in the same way that the electron energy is measured upward from the bottom. If a band is less than half-filled we usually think in terms of the electrons as the charge carriers. For bands that are more than half-filled, some of the electron states are not paired in momentum space and we usually think in terms of the holes as carriers.

THE HALL EFFECT

In 1879 E. H. Hall[5] observed that an emf may be generated across a current-carrying conductor by placing the conductor in a magnetic field. When the direction of the current flow is perpendicular to the direction of the magnetic field, a voltage will be developed

[5] E. H. Hall, ***Am. J. Math., 2, 287 (1879)***

across the sample at right angles to both the current and the magnetic field. This voltage is known as the ***Hall voltage.***

An electric field applied to the sample along the x-direction gives rise to a current density $\mathbf{j}_x$. As the charge carriers move through the sample they are deflected by the magnetic field $\mathbf{B}_z$. In the steady state there will be no current in the y-direction. This is accomplished by a concentration gradient of carriers which produces an electric field to oppose the magnetic deflection. The force on a carrier of charge e and velocity v is just the Lorentz force.

$$\mathbf{F} = e[\mathcal{E} + \mathbf{v} \times \mathrm{B}] \tag{5-39}$$

The steady state condition demands that $F_y = 0$. This means that

$$e\mathcal{E}_y - ev_x B_z = 0 \tag{5-40}$$

for a carrier velocity v_x and a geometry shown in Fig. 5-9. Using the

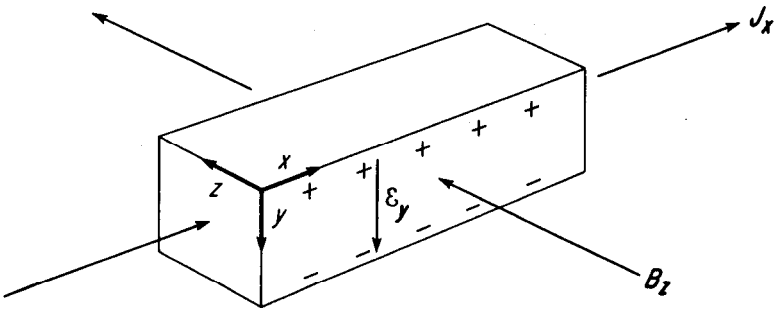


Fig. 5-9. Hall-effect geometry. Current in x-direction, magnetic field in z-direction, and Hall voltage in y-direction.

relation for current density

$$j_x = ne\, v_x \tag{5-41}$$

the Hall field $\mathcal{E}_y$ may be expressed as

$$\mathcal{E}_y = \frac{j_x B_z}{ne} \tag{5-42}$$

Upon dividing $\mathcal{E}_y$ by $j_x B_z$ we get the Hall coefficient

$$R = \frac{\mathcal{E}_y}{j_x B_z} = \frac{1}{ne} \tag{5-43}$$

The Hall coefficient is seen to be inversely proportional to the number of charge carriers per unit volume, often called the carrier concentration. In addition, the sign of the Hall coefficient reflects the effective sign of the charge carrier. For the alkali metals and the noble

metals, experimentally observed Hall coefficients are close to the values calculated from the free electron theory as seen in Table 5-1.

Table 5-1
Ratio of Observed Hall Coefficients to Free Electron Model Calculations (after Ziman*) n = Atomic Concentration

Metal	$R_{exp} \sim ne$	Metal	$R_{exp} \sim ne$
Li	-1.3	Be	+5
Na	-0.9	Zn	+0.4
K	-0.9	Cd	+0.5
Rb	-1.0	Ca	-0.7
CS	-1.1	Al	-0.4
cu	-0.8	Cr	+5
Ag	-0.8	Fe	+14
Au	-0.7	Co	+4

*J. M. Ziman, ***Electrons and Phonons*** (Oxford: Clarendon Press, 1970), p. 488.

For many of the other metals the agreement is poorer and, as in the case of beryllium, there is a reversal of the sign of the effective charge carrier. This reversal of sign of the Hall effect and the influence of the periodic potential upon the effective mass of the electron forms the basis for describing conduction in terms of both electrons and holes.

Sometimes one encounters an additional factor of $3\pi/8$ in the expression for the Hall constant. This factor arises from the choice of a model in which the mean free path of the charge carrier is independent of energy and in which the relaxation time depends on energy through $\tau \propto E^{1/2}$. This model is frequently appropriate for semiconductors where the Fermi surface is essentially spherical and the dominant scattering is acoustic-mode lattice scattering. However, this correction factor is usually not important in comparison with other errors in the calculation.[6]

The Hall constant given above (Eq. 5-43) is appropriate only for one type of carrier. If two types of carriers are present, for example both electrons and holes in a semiconductor, an analysis similar 'to that above leads to

$$R = \frac{1}{e} \frac{p\mu_p^2 - n\mu_n^2}{(p\mu_p + n\mu_n)^2} \tag{5-44}$$

[6] E. H. Putley, ***The Hall Effect and Related Phenomena*** (London: Butterworth, 1960).

where p and n are the concentrations of holes and electrons. μ_p and μ_n are the mobilities of the holes and electrons. In the event that one type of carrier is especially dominant, Eq. 5-44 reverts to the original expression of Eq. 5-43.

CLASSIFICATION OF CRYSTALS

We have already seen that a collection of N regularly spaced potential wells possesses distinct sets of quasi-continuous energy levels called *bands.* Furthermore, these bands are separated by regions of energy inaccessible to the electrons and thus referred to as forbidden zones. Each allowed band is characterized by a set of N distinct k-values and thus N closely spaced energy levels. In addition, there is the twofold degeneracy arising from the electron spin. Therefore, each energy band contains 2N independent states.

Moreover, we showed earlier that a completely filled band did not contribute to conduction any more than a completely empty band. We might expect that the best conductors would be those which have a half-filled allowed band. Such is the case for the alkali metals. As an example, sodium has a single 3s electron above a core of filled levels. Thus the energy band in solid sodium which corresponds to the 3s atomic level is only half-filled. We find that sodium has a high conductivity and is considered to be a good metal.

We might expect that elements in the second column of the periodic table would be insulators because each atom would contribute two electrons to the highest occupied band. This is not the case for they are metals although their conductivities are not so good as those of the first column elements. Their conductivities arise from

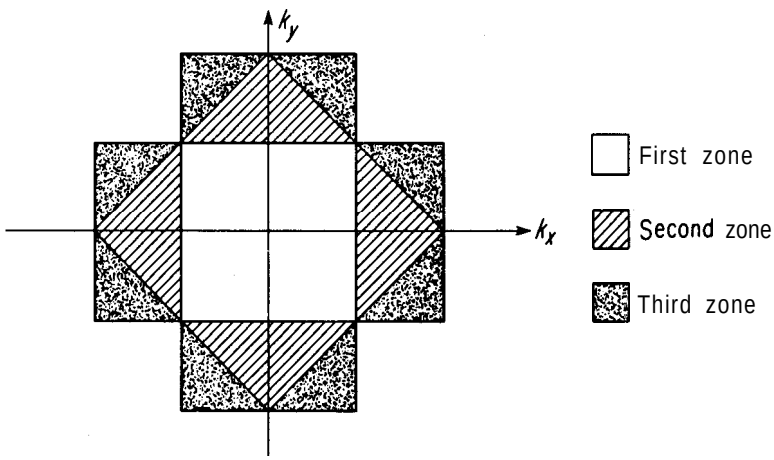


Fig. 5-10. Brillouin zones of a simple square lattice in two dimensions. The first three zones are shown.